

Sphere interactions (Yukawa, polynomial, general)

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June 2026

Introduction

A bit late, but this has been bothering me for ages, so I'm finally closing the loop. The main new derivation here is the final expression 7. Maybe this can help prepare for a non-Yukawa dependent limit plot...

Potential on the sphere

For an arbitrary interaction between a sphere of density ρ and a point mass m characterized by a radially symmetric potential $m\rho V(r)$

$$V(r', r_b, \rho) = m \int_{r'-r_b}^{r'+r_b} \int_0^\pi \int_0^{2\pi} \rho V(r) r^2 \sin(\theta) d\phi d\theta dr \quad (1)$$

The geometry of this setup is shown in Figure 1, where s is the separation between the point mass and the closest point on the surface of the sphere, r_b is the radius of the sphere, and $r' = s + r_b$ is the distance to the center of the sphere. Symmetry can be used to perform the integration over the θ and ϕ coordinates and sum up spherical caps weighted by $V(r)$.

The area of the spherical cap intersecting the sphere at a distance r from the point mass is given by

$$A(r) = r^2 \int_0^{\theta_{\max}} \int_0^{2\pi} \sin(\theta) d\phi d\theta = 2\pi r^2 (1 - \cos(\theta_{\max})) \quad (2)$$

where $\theta_{\max}$ is the maximum azimuthal angle at which the spherical cap centered on the point mass intersects the sphere. $\theta_{\max}$ can be calculated from the law of cosines as

$$\theta_{\max} = \arccos\left(\frac{(r')^2 + r^2 - r_b^2}{2rr'}\right) \quad (3)$$

Inserting this into Equation (3), we obtain

$$A(r, r') = \pi r \frac{r_b^2 - (r + r')^2}{r'} \quad (4)$$

So we can write the total potential between a point mass and a sphere as

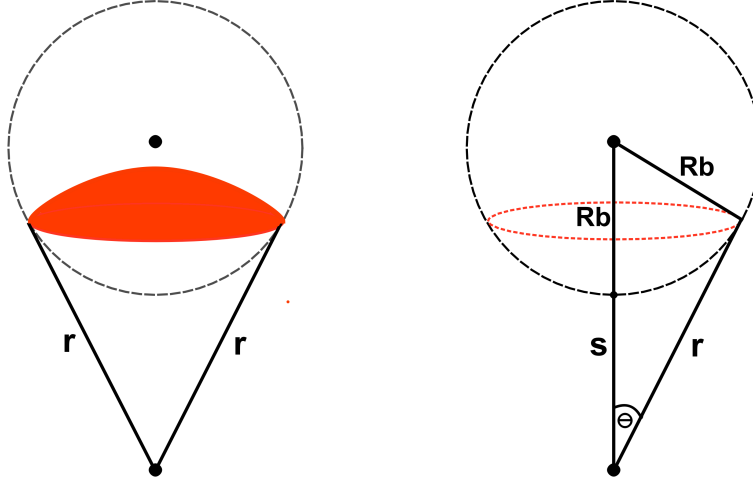


Figure 1: The geometry for integrating a function that is constant of radius between a point and a sphere.

$$V(r', r_b, \rho) = m\rho \int_{r'-r_b}^{r'+r_b} V(r) A(r, r') dr \quad (5)$$

The total force on the sphere must therefore be

$$F(r', r_b, \rho) = -\hat{\mathbf{r}}' \frac{d}{dr'} m\rho \int_{r'-r_b}^{r'+r_b} V(r) A(r, r') dr \quad (6)$$

Applying Leibniz rule and noting that $A(r, r')$ vanishes at the integral endpoints,

$$F(r', r_b, \rho) = -\hat{\mathbf{r}}' m\rho \int_{r'-r_b}^{r'+r_b} V(r) A'(r, r') dr \quad (7)$$

where

$$A'(r, r') = \frac{\partial}{\partial r'} A(r, r') = \pi r \frac{r^2 - r_b^2 - (r')^2}{(r')^2} \quad (8)$$

What happens if we integrate force instead of potential

Say we want to integrate the force first. Then, we want

$$\vec{F} = \iiint -\nabla V(r) dv \quad (9)$$

Since V is a radially symmetric potential, the azimuthal components of the gradient vanish, and we are left with

$$\vec{F} = \iiint -\frac{\partial V}{\partial r} \hat{r} dv \quad (10)$$

Following the same spherical cap method, we pull out the radially dependent terms (assuming constant density), leaving

$$\vec{F} = - \int_s^{s+2r_b} r^2 \frac{\partial V}{\partial r} \int_0^{\theta_{max}} \int_0^{2\pi} \sin(\theta) \hat{r} d\phi d\theta dr \quad (11)$$

Now, appealing to symmetry, any component of $\hat{r}$ that is not parallel to a line connecting the point mass to the center of the sphere will cancel in the integral. So, we can simply integrate the scalar dot product of $\hat{r}$ with this line, which we denote $\hat{r}'$. By the definition of our angles, this dot product is given by $\cos \theta$, so the integral becomes

$$\vec{F} = -\hat{r}' \int_{r'-r_b}^{r'+r_b} r^2 \frac{\partial V}{\partial r} \int_0^{\theta_{max}} \int_0^{2\pi} \sin(\theta) \cos(\theta) d\phi d\theta dr \quad (12)$$

$$\vec{F} = -\hat{r}' \int_{r'-r_b}^{r'+r_b} r^2 \frac{\partial V}{\partial r} \pi (1 - \cos^2(\theta_{max})) dr \quad (13)$$

$$\vec{F} = -\hat{r}' \int_{r'-r_b}^{r'+r_b} r^2 \frac{\partial V}{\partial r} \pi \left(1 - \left(\frac{(r')^2 + r^2 - r_b^2}{2rr'} \right)^2 \right) dr \quad (14)$$

Performing integration by parts with the boundary terms canceling recreates equations 7 and 8, and so the order of integration is irrelevant.

Summary of numerical results

Newtonian gravity:

$$V(r) = -\frac{Gm_1m_2}{r}$$

$$F_{grav} = -\frac{Gm\rho}{r^2}v(r_b)$$

where

$$v(r) = \frac{4}{3}\pi r^3$$

Yukawa potential:

$$V(r) = -\frac{Gm_1m_2}{r}(1 + \alpha e^{-r/\lambda})$$

$$F_{yuk} = F_{grav} - \frac{Gm\rho}{r} \left(\frac{3}{2} \alpha e^{-s/\lambda} v(\lambda) f(r_b/\lambda) \right) \left(\frac{1}{r} + \frac{1}{\lambda} \right)$$

where

$$f(r_b/\lambda) = e^{-2r_b/\lambda}(1 + r_b/\lambda) + r_b/\lambda - 1$$

Polynomial potential:

$$V(r) = -\frac{Gm_1m_2}{r}(1 + \alpha(\frac{r_0}{r})^{N-1})$$

$$F_{N=2} = -\frac{Gm\rho\pi\alpha x_i}{r^3}(2r_b r + (\log(s))(r_b^2 + r^2) - (\log(r_b + r))(r_b^2 + r^2))$$

$$F_{N=3} = -\frac{Gm\rho\pi\alpha x_i}{r^3}(\frac{r_b^3}{r_b^2 - r^2})$$

$$F_{N=4} = -\frac{Gm\rho\pi\alpha x_i}{r^3}(-\text{ArcTanh}(\frac{r_b}{r}) + \frac{r_b r(r_b^2 + r^2)}{(r_b^2 - r^2)^2})$$

$$F_{N=5} = -\frac{Gm\rho\pi\alpha x_i}{r^3} \frac{4r_b^3(r_b^2 - 5r^2)}{3(r_b^2 - r^2)^3}$$

Kernel for arbitrary potential

We can take this to its logical conclusion by calculating the full force from the entire attractor. Technically this should be

$$\vec{F}_{tot} = \iiint -\hat{\mathbf{r}}' m(\mathbf{r}') \rho \int_{r'-r_b}^{r'+r_b} V(r) A'(r, r') dr d\mathbf{r}' \quad (15)$$

but we usually approximate as a sum of point masses

$$\vec{F}_{tot} = \sum_{\mathbf{r}'} -\hat{\mathbf{r}}' m(\mathbf{r}') \rho \int_{r'-r_b}^{r'+r_b} V(r) A'(r, r') dr \quad (16)$$

Reversing the order of summation and integral gives

$$\vec{F}_{tot} = \int_0^\infty dr V(r) \sum_{\mathbf{r}': r-r_b < r' < r+r_b} \hat{\mathbf{r}}' \rho m(\mathbf{r}') A'(r, r') \vec{F}_{tot} = \int_0^\infty V(r) \vec{K}(r) dr \quad (17)$$

We want to do this in frequency domain. It's safe to assume that the potential is time independent, and so the only explicit time dependence is $\vec{K}(r, t)$, which comes from changing $\mathbf{r}'$ over time. As such, taking the fourier transform is simply a matter of computing $\vec{K}(r, \omega)$ and plugging into 17, giving

$$\vec{F}_{tot}(\omega) = \int_0^\infty V(r) \vec{K}(r, \omega) dr \quad (18)$$

Thus, the complex kernel $\vec{K}(r, \omega)$ gives our sensitivity at all length scales. This can be calculated directly from geometry using the same FEA analysis as previously implemented for our templates.